Detecting Cyber Threats at Scale:

Anomaly Detection in Network and System Logs Using Machine Learning

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Introduction

Types of Security Logs | A Complete Guide for Cybersecurity Monitoring in 2026

Cyber threats don't announce themselves, they hide in plain sight, buried within massive streams of everyday system activity. Every time a user logs in, accesses a file, sends a request, or interacts with an application, that action is recorded as a security log. These logs are generated by systems, applications, and network devices, and they track important events such as login attempts, access control changes, network traffic, and suspicious behavior . There are multiple types of logs including system logs, application logs, network logs, security logs, transaction logs, and audit logs, all of which together create a detailed digital footprint of how a system operates. While these logs are incredibly valuable, they are also overwhelming in scale, with modern systems producing millions or even billions of entries daily. This project focuses on using machine learning to automatically analyze these logs and detect unusual patterns that may indicate cyber threats. At a big-picture level, the goal is to turn raw, complex data into actionable insights that help identify attacks early and strengthen overall security.

This work is especially relevant in today's society, where nearly every aspect of life like banking, healthcare, education, communication, and online platforms are dependent on secure digital systems. As cyber attacks grow more advanced, organizations can no longer rely solely on traditional, rule based detection methods. These systems often fail to catch new or evolving threats, leaving sensitive data and user privacy at risk. Security logs play a critical role in modern cybersecurity because they act as the "source of truth" for identifying suspicious activity, enabling organizations to detect insider threats, unauthorized access, and system breaches . However, without intelligent tools, the sheer volume of log data makes timely detection nearly impossible. This project demonstrates how machine learning can shift cybersecurity from a

reactive approach to a proactive one, allowing organizations to detect threats earlier, respond faster, and ultimately build more trustworthy digital environments.

The problem this project addresses is the difficulty of identifying rare and hidden threats within vast amounts of complex log data. Manual monitoring is not scalable, and traditional systems struggle to detect subtle or previously unseen attack patterns. This project solves that problem by applying both unsupervised and supervised machine learning techniques to automatically identify anomalies, or unusual behaviors that may signal malicious activity. It is designed to support cybersecurity analysts, IT teams, and organizations that rely on secure systems to protect sensitive information. By improving the accuracy and efficiency of threat detection, this project has real world impact: it helps prevent data breaches, reduces financial loss, and protects user privacy. Ultimately, it showcases not only technical skill, but also the ability to build meaningful solutions that contribute to a safer and more secure digital society.

Machine Learning Topics

A full breakdown of machine learning topics can be located on the CS406 Project Repo Wiki.

The objectives of this project are to apply the full data science and machine learning workflow to a real world dataset while demonstrating competency in data preparation, unsupervised learning, supervised modeling, and comparative model evaluation. Specifically, the project will:

- Perform Data Gathering
- Conduct Data Cleaning and Preparation
- Apply Unsupervised Learning Techniques (Clustering and PCA)
- Implement Linear Regression Modeling
- Implement Logistic Regression Modeling
- Apply Decision Tree and Tree-Ensemble Classification Methods
- Implement Bayesian Classification Techniques
- Apply Support Vector Machine Classification
- Implement Ensemble Classification Methods
- Compare and Interpret All Modeling Approaches

Any additional objectives met/trialed will be added in this section.

Links to Artifacts

- Public Github Repo
- Original Proposal
- Machine Learning Topics
- Term Timeline

Thinking Questions

The following questions are designed to guide this project beyond simply building and evaluating machine learning models, but instead focus on understanding their real world impact and usefulness. While technical performance is important, the true value of this system lies in how effectively it can detect threats, support decision making, and operate within real cybersecurity environments. These questions aim to explore not only how well the models perform, but also what their results mean in practical terms, such as how they handle risk, how they would be used by analysts, and how they could be integrated into existing systems. By answering these questions, the project evaluates both the technical success and the real world applicability of machine learning for cybersecurity anomaly detection

1. Can this system detect cyber threats early enough to prevent real damage?:
    a. This question evaluates whether the model can identify suspicious activity quickly enough to allow organizations to respond before a breach or attack causes harm.

2. What is the acceptable balance between false positives and false negatives?
    a. This explores the trade-off between flagging too many normal events as threats versus missing actual attacks, which directly impacts system usability and risk.

3. Which types of anomalies or attacks are easiest and hardest to detect?
    a. This helps identify where the model performs well and where it struggles, revealing limitations in detecting subtle or complex threats.

4. Can unsupervised learning detect previously unseen or evolving cyber threats?
    a. This examines whether clustering and PCA can uncover new attack patterns that were not included in the training data.

5. Which machine learning model provides the best performance for anomaly detection?
    a. This compares different models to determine which approach is most effective based on accuracy, precision, recall, and overall reliability.

6. How well do the models generalize to new, unseen data?
    a. This evaluates whether the trained models can maintain performance when applied to different datasets or real-world environments.

7. How does dimensionality reduction (PCA) impact anomaly detection performance?
    a. This investigates whether reducing feature complexity improves model efficiency and the ability to detect meaningful patterns.

8. How can this system be integrated into a real-world cybersecurity workflow?
    a. This considers how the model's outputs could be used within tools like SIEM systems to support real-time monitoring and decision-making.

9. How would cybersecurity analysts interpret and act on the model's results?
    a. This focuses on the usability of the system and whether its outputs provide clear, actionable insights for human users.

10. What impact could this system have on improving cybersecurity in today's digital world?
    a. This explores the broader significance of the project, including its potential to protect sensitive data, reduce risk, and strengthen trust in digital systems.

Coming Soon

This section will serve as a placeholder for topics in order to comply with requirements for Project 1. This section is subject to deletion upon the completion of this project.

The Datasets

Data Cleaning, Preparation, and EDA

Unsupervised Learning (Clustering and PCA)

Regression Models

Classification Models Part 1:

Classification Models Part 2:

Analyses

Phase 1

Phase 2

Phase 3

Results

Conclusions

Project Sources

(N.d.). Retrieved from https://www.webasha.com/blog/types-of-security-logs-a-complete-guide-for-cybersecurity-monitoring